PerkinElmer UV/VIS/NIR SOP Last Updated: 9/19/2022

1) Use the windows login UVvisNIR_USER
    a. no password required
2) Launch from the Desktop **PerkinElmer UV WinLab**
    a. Choose **analyst** as the user
3) On the left-hand group of icons choose **Scan- Lambda 750**
    a. Ignore the halogen lamp error that pops up
4) If necessary (software will prompt you with a pop-up dialog box), do the wavelength calibration
    a. This is done automatically by the software with no samples in the spectrophotometer
    b. When it finishes, it should say calibration complete, then click the OK box to dismiss the pop dialog box
5) Under **Data Collection** enter your data collection parameters
6) Under **Corrections** select
    a. Baseline correction: always before the next measurement
    b. Check the 100% T box
    c. Check the 0% T box
    d. (if you have a lot of samples to run under the same experimental conditions you can uncheck these after the first sample is finished)
7) Under **Sample Info** enter your sample name
8) Press **Start** icon
    a. First it will do the 100%T correction, with no samples in the compartment
    b. Next it will do the 0%T correction, with the black plastic cuvette (located in the top drawer of the desk) in the front cuvette holder inside the spectrophotometer
    c. Now run your sample, sample in the front cuvette holder, solvent in the back cuvette holder. Use only quartz cuvettes.
9) To save, FILE -> Save Spectra
    a. Save in My Documents in a folder you create
10) To export, FILE -> Export
    a. In dialog box, change path to save to My Documents and your folder
    b. Check the box near XY Data (raw) Samples by **Columns**

Optical Spectroscopy — Session 01

Document 3 of 3 — PerkinElmer Lambda 750 UV/Vis/NIR Spectrophotometer

UNR Shared Instrumentation Laboratory, Room 016 | Monday, April 13, 2026

Third and final instrument — read documents 1 and 2 first

This is document 3 of 3 for Monday's session. Document 1 (UV-2550 Protocol) is the master document with all sample preparation and packing. Document 2 (FluoroMax-3) is the fluorescence session. This document picks up immediately after the FluoroMax session ends.

By the time you reach this bench you should be carrying: the six sample stock bottles, both blank bottles (H2O, EtOH), clean cuvettes, wash bottles, waste bottle, and a USB stick with the UV-2550 and FluoroMax-3 data already on it.

Session sequence — you are at the last stop

1. UV-2550 (doc 1) — 09:30 – 10:15. ✓ completed.

2. FluoroMax-3 (doc 2) — 10:30 – 11:45. ✓ completed.

3. Lambda 750 (THIS document) — 12:00 – 13:00. Cross-validate the UV-2550 data on a research-grade double-beam instrument, AND extend the scan range into the near-infrared (800 – 2500 nm) to see features neither the UV-2550 nor the FluoroMax can reach.

Lambda 750 session goal

Two jobs: (1) cross-validation — rescan the same six samples on the Lambda 750 from 200 – 800 nm and compare the spectra against the UV-2550 results. A research-grade double-beam double-monochromator instrument should agree with the UV-2550 to within a few percent on A values and a nanometre on peak positions. Any large disagreement is a calibration or cleanliness flag.

(2) NIR extension — rescan both blanks (H2O and EtOH) from 800 – 2500 nm. The near-infrared is where water shows broad O-H vibrational overtones (~970, 1200, 1450, 1940 nm) and ethanol shows C-H overtones (~1400, 1700 nm). These are completely invisible to the UV-2550. Lambda 750's PbS detector is what lets us see them.

Expected duration: 60 minutes including warmup.

Key concept — why the NIR looks different from the UV/Vis

UV/Vis absorption (200 – 800 nm) probes ELECTRONIC transitions — electrons promoted from ground-state orbitals to excited-state orbitals. That is what gives fluorescein its color.

NIR absorption (800 – 2500 nm) mostly probes VIBRATIONAL OVERTONES and COMBINATIONS — specifically overtones of O-H, N-H, and C-H stretches. These are weak (they are formally forbidden transitions), but they are spectacularly useful because every molecule with O-H or C-H bonds shows them at characteristic wavelengths.

So: dye molecules show strong features in the UV/Vis but weak or none in the NIR. Pure solvents show nothing in the UV/Vis (that's why you use them as blanks) but become INTERESTING in the NIR. The two regions are complementary.

Safety reminder — Lambda 750 specific

- The Lambda 750 uses a deuterium lamp (UV) and a tungsten-halogen lamp (Vis/NIR). Both are UV/bright sources. Do not stare into the sample compartment with the lid open.
- The Lambda 750 has a cooled (Peltier or liquid-N2) PbS detector for the NIR. If a liquid-N2 dewar is mounted, do NOT tilt or move the instrument. If you are unsure whether the detector is cooled and ready, check the software status indicator.
- Full PPE, flammables, and waste handling: see the master safety checklist in document 1.

Part A — Samples (see master document)

Samples are the same six fluorophore stocks + two blank solvents prepared on Sunday evening per document 1 (UV-2550 Protocol), Part A. No additional prep. Use the ORIGINAL undiluted stocks, not the dilutions you made for the FluoroMax session.

Quick sanity check before starting: all six amber bottles present, both blanks present, cuvettes clean, waste bottle has capacity.

Part B — Lambda 750 warmup and setup

The Lambda 750 is a more sophisticated instrument than the UV-2550: double-beam, double-monochromator, and a second detector for the NIR region. It still warms up reasonably quickly.

Step B1 — Power on

1. Turn on the main power switch. You will hear the cooling fan spin up and the lamp ignition click.
2. Start UV WinLab (the PerkinElmer software) on the instrument PC. It will initialize and do a self-test of wavelength drive, detector changeover, and grating changeover. Let it finish.
3. Start a 15-minute warmup timer. The D2 and tungsten lamps both need thermal stabilization; a cold lamp drifts during a scan.
4. Check the PbS NIR detector status in the software. If it reads "cooling" or "not ready", wait until it says "ready".

Step B2 — Cuvette preparation

Same ritual as the UV-2550. For NIR scans that extend to 2500 nm, use the same quartz cuvettes — standard UV-grade fused silica is transparent from ~170 nm out to ~2700 nm, comfortably covering the Lambda 750's full range.

5. Clean both cuvettes as in document 1 Part C: 3× water, 1× ethanol, 1× water, Kimwipe polish.
6. Inspect the two polished optical faces. No fingerprints, no streaks.

Step B3 — Software configuration

Parameter	Value	Why
Measurement mode	Scan	Wavelength sweep
Ordinate	Absorbance (A)	Same as UV-2550 for direct comparison
UV/Vis scan range	200 – 800 nm	Matches the UV-2550 range for cross-validation
NIR scan range	800 – 2500 nm	Full NIR reach of the PbS detector
Data interval	1.0 nm (UV/Vis), 2.0 nm (NIR)	NIR features are broader; 2 nm is plenty
Scan speed	Medium (~400 nm/min)	Balance speed and SNR
UV/Vis slit	2.0 nm	Matches UV-2550 setting for comparability
NIR slit	Servo (auto-track)	Lambda 750 auto-adjusts for PbS sensitivity
Lamp changeover	Auto at ~320 nm	Default D2 → tungsten handoff
Detector changeover	Auto at ~860 nm	PMT → PbS handoff
Grating changeover	Auto at ~860 nm	UV/Vis grating → NIR grating
Data export	CSV per spectrum	Loadable in Python

Notes about the changeovers: when the instrument crosses the lamp / detector / grating boundaries during a scan you may see a small step or kink in the spectrum. That is normal. Research-grade post-processing removes it; for qualitative work, just ignore.

Step B4 — Baseline correction

Two baselines: one for the UV/Vis range and one for the full UV/Vis/NIR range (needed for the blank-solvent NIR scans).

7. Fill both cuvettes with distilled water. Place one in the sample position, one in the reference position.
8. Run a 200 – 800 nm baseline. Save as baseline_UVVIS_H2O.
9. Now run a 200 – 2500 nm baseline with the same water cuvettes. Save as baseline_FULL_H2O. This one takes ~3 minutes.
10. You will re-baseline later (fresh ethanol in both cuvettes, 200 – 2500 nm) when you switch to ethanol samples.

Part C — Scan plan (two halves)

Part C1 is the cross-validation half: six samples, 200 – 800 nm, directly comparable to the UV-2550 data. Part C2 is the NIR half: two blank solvents pushed out to 2500 nm plus one or two samples for a bonus look.

C1 — Cross-validation sweep (200 – 800 nm)

Use the same undiluted stock that gave a clean A on the UV-2550 (if UV-2550 reported A > 1.5 for a stock, use a 1:1 dilution here). Goal: one spectrum per sample, directly comparable to the UV-2550 spectrum.

11. Order: quinine, yellow HL, pink HL, salicylate (all in water, water baseline), then re-baseline with ethanol, then curcumin, green tea.
12. One spectrum per sample, same file-naming convention: 20260413_Lambda750_UVVIS_S{n}_{sampleName}.csv
13. At the end, open the Lambda 750 and UV-2550 CSVs side by side in your notebook and visually confirm the peak positions agree within ~1 nm. Large discrepancies flag a wavelength calibration issue on one of the two instruments.

C2 — NIR extension (800 – 2500 nm)

The interesting NIR features are in the solvents, not the dyes. Priority scans:

Scan	Sample	Range	Expected NIR features
NIR-1	Distilled water (blank)	800 – 2500 nm	Water O-H overtones at ~970, ~1200, ~1450, ~1940 nm
NIR-2	95% ethanol (blank)	800 – 2500 nm	C-H overtones at ~1400, ~1700 nm + residual water O-H
NIR-3 (bonus)	Green tea / EtOH extract	800 – 2500 nm	Same ethanol features + chlorophyll weak features
NIR-4 (bonus)	Curcumin / EtOH	800 – 2500 nm	Same ethanol background + any curcumin NIR tail

NIR scans are the pedagogical centerpiece of the Lambda 750 session. Point out to Vivian: the ~1940 nm water peak is massive — water is nearly opaque in this region. This is why the entire astronomical atmosphere has an "opacity window" structure, and why IR telescopes go to high dry sites or space.

Part D — Standard procedure per sample

14. Rinse the sample cuvette once with the new sample (pre-rinse); discard into waste bottle.
15. Fill the sample cuvette with ~3 mL of sample. Leave the reference cuvette with the active blank.
16. Wipe both polished faces with a fresh Kimwipe. Inspect. Place in the sample position.
17. Run the scan (either 200 – 800 nm for C1 or 200 – 2500 nm for C2).
18. Save as CSV with the appropriate filename.
19. Between-sample cleaning: 3× water, 1× ethanol, 1× water, Kimwipe polish.

Part E — Shutdown and packout

20. Verify all CSVs are saved. Open two at random to confirm they parse.
21. Copy the Lambda 750 folder to the USB stick.
22. Close UV WinLab.
23. Turn off the Lambda 750 at the main power switch. Do NOT unplug — fan may run briefly.
24. Wipe down the benchtop with a fresh Kimwipe.
25. NOW pack out everything you brought in across the whole morning: all cuvettes, tweezers, pipettes, sample bottles, blank bottles, wash bottles, waste bottle, printed protocols, USB stick, phone. Double-check the bench is empty. 016 permits no storage.
26. Log your visit in the SIL access book (time in, time out, instruments used).

Part F — Data management

27. Copy the session folder into GITHUB/science/research/20260413 Lambda-750/raw/
28. Create an index.md with: date, instrument (PerkinElmer Lambda 750), operators, samples, file list, and a short note about any observed disagreement with the UV-2550 cross-validation.
29. Load the water NIR spectrum in Python and save a PNG with the four overtone peaks labeled (~970, 1200, 1450, 1940 nm). This is the headline figure for the Lambda 750 session.
30. In a single commit, add all three sessions' data: UV-2550, FluoroMax-3, and Lambda 750. Commit message: "Add Session 01 optical spectroscopy — 3 instruments, 6 fluorophores, cross-validation + NIR".
31. Bonus: create a cross-instrument comparison plot with the UV-2550 and Lambda 750 quinine spectra overlaid. Same file, two traces, different colors. That plot is the scientific payoff of Monday's entire morning.

Part G — Troubleshooting cheat sheet

Symptom	What to try
UV/Vis spectra disagree with UV-2550 by more than 5 nm on peak position	Both instruments should be within 1 nm. Suspect a wavelength calibration issue. Check the Lambda 750's holmium oxide calibration if available. Re-scan a clean water blank to confirm the baseline.
A step or kink at 320 nm or 860 nm	Lamp or detector changeover — normal. Does not affect peak positions.
NIR spectrum shows strong noise above 2200 nm	PbS detector approaching its long-wavelength limit. Widen data interval to 4 nm or accept the noise.
NIR baseline drifts upward with time	Room temperature or humidity change affecting water vapor in the optical path. Re-baseline.
Peak at 1450 nm saturates (A > 3)	Water O-H overtone is naturally very strong. Use a shorter path length (2 mm cuvette if you have one) — or accept the saturated peak for a qualitative view.
Two samples look identical in the UV/Vis	Good. They should. You are looking at the same dye in the same solvent. Check they differ in the visible region if you expected them to.
All A values uniformly low vs UV-2550	Cuvette mismatch: the Lambda 750 reference cuvette may have a different path length. Use matched pair cuvettes.
Scan takes much longer than expected	Data interval too small (0.2 nm) or scan speed too slow. Both are adjustable; Medium / 1 nm is the default.

Part H — Post-session reflection (the big synthesis)

These are the questions that make the whole three-instrument morning worth it. Discuss with Vivian on the walk back to the car.

- For quinine, you now have (a) its absorption spectrum from the UV-2550, (b) its emission and excitation spectra from the FluoroMax-3, and (c) its absorption spectrum AGAIN from the Lambda 750. Which data set was the easiest to collect? Which was the most informative? Which was the most demanding of sample prep?
- The Lambda 750 and UV-2550 should agree closely on the quinine absorption peak. If they do, what does that tell you about instrument calibration in general? If they don't, which do you trust more, and why?
- The water NIR spectrum has four clearly resolved overtone bands. How do you identify them physically — what is the underlying vibration, and why are there four (not one)?
- Ethanol has both O-H and C-H bonds, so it should show BOTH water-like features and additional C-H features. Can you tell them apart in the spectrum?
- Based on what you saw this morning, design a Session 02: what is one follow-up experiment you could do with JUST ONE of these instruments that would turn the qualitative morning into a quantitative result?

This is what real research looks like: the first morning is always qualitative and exploratory. The numbers come second. The questions come third. Pick the question Vivian finds most interesting and build Session 02 around it.

— End of Session 01 — three instruments, six samples, one good morning —